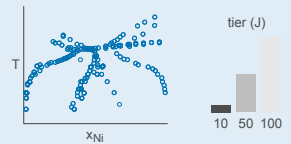
ORCHID-SAGE

a CALPHAD assessment recorded step by step, so that another machine regenerates it (Cr–Ni)

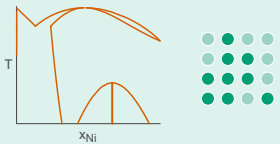
ORCHID unreleased

1 Evidence



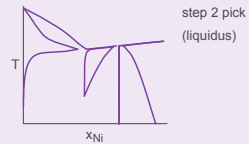
647 ORCHID states: id, σ
318 with excluded phases
certainty tiers

2 One-shot



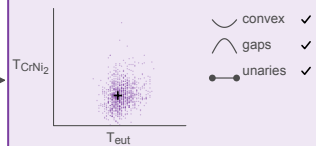
PhaseForge MLIP, m_scraps SQS
staged relaxation, SGTE refs
100 ps liquid MD, S_{ex} corrected

3 Fit



PhaseForge+ gradient ESPEI
liquidus → thermo → solid
hinge; topology-first ranking

4 Checks and UQ



convexity, gaps, unary limits
Laplace mixture, 1000 draws

Record

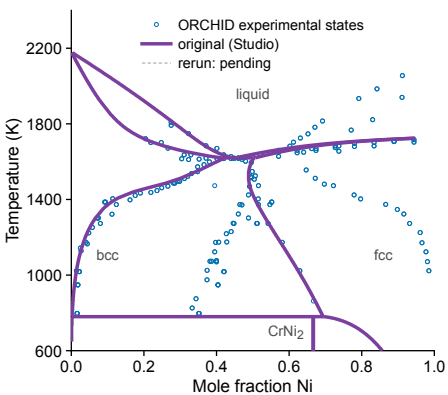
state ids | sha256 | image digest | seeds | commit, lock | trajectory | draws | decisions

People decide
8 decisions logged: who, why, source

Package

- specific protocol
- Workflow Run RO-Crate
- dashboard
- TDB + header

Machine B reads only the package and returns the same diagram



step	person	agent
one-shot TDB, step-1 base	pending	pending
evidence bundle, TDB header	pending	pending
MLIP energies (GPU)	pending	pending
step 2 liquidus	pending	pending
step 3b thermo	pending	pending
step 4 pick TDB	pending	pending
eutectic, $CrNi_2$ top	pending	pending

= byte-identical ≈ within tolerance × not reproduced

person
`orchid reproduce`
rerun: pending

AI agent
ORCHID-SAGE MCP
`read protocol` `run steps` `compare`
rerun: pending (MCP not built)

Evidence class ○ Experimental ■ MLIP / DFT — CALPHAD model — ORCHID-SAGE fit and record